Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for small retail and service businesses, turning expansion questions into site-selection analyses; identified a $1.5M expansion opportunity.
- Estimated the incremental revenue of new locations with Double ML and causal forests (econml, XGBoost nuisance models) on a 12-year county panel, validated by DiD and a synthetic-data recovery test.
- Built Monte Carlo financial risk models (probability of loss, VaR, CVaR) from client operating costs, local demand, and the causal estimates' 95% confidence intervals.
- Engineered Python ETL pipelines with automated data-quality checks and 150+ unit tests for Census, traffic, and competitor data, cutting market analysis turnaround by 91% (weeks to days).
- Designed a site-scoring model combining demographic, traffic, competitive-density, and consumer-behavior features, presenting results to non-technical owners as written reports and maps.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap with strategy, engineering, and behavioral science leadership, setting OKRs and translating business questions into models.
- Applied predictive modeling, customer segmentation, and uplift modeling to millions of Medicare/Medicaid-eligible consumer records, reducing ad spend by $10K per campaign.
- Designed and standardized cross-team A/B testing and causal inference frameworks for direct-mail campaigns, applying CUPED to reduce metric variance by 20%.
- Owned a monthly model monitoring report tracking drift and performance for up to 10 production models, including media mix, response, eligibility, and psychology-match models.
- Re-architected a production churn prediction platform in Python, SQL, Spark, Snowflake, and Databricks, reducing runtime by 48% and memory consumption by 23%.

Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- Defined success metrics, evaluation frameworks, and drift monitoring for production search and regression models using offline validation and online experimentation.
- Built SQL training tables grouping bulk orders into item-level shipping costs, then deployed a regression-based shipping-price API in C# on Azure, scaling to 10,000+ daily requests.
- Reduced search customer dissatisfaction by 12% with product and engineering partners by building NLP relevance features: custom Elasticsearch indexing, tokenizers, and TF-IDF weighting.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Built and validated XGBoost classification models on millions of healthcare consumer records with hundreds of external attributes, driving a 3% lift in response rates.
- Automated reporting pipelines and Power BI dashboards for stakeholders with Python, SQL, and Snowflake, removing 60 minutes of manual work daily.
- Improved geospatial targeting accuracy by 20% through GIS analysis with GeoPandas, focusing campaign resources on high-opportunity regions.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in mathematical optimization, functional analysis, and operator theory for doctoral dissertation research.
- Taught mathematics and statistics to classes of 75-200 students and mentored 100+ students in summer programs, achieving a 95% pass rate.

TECHNICAL PROJECTS

CaLaun — *AI-Powered Calculus Tutor (LLM Application)* github.com/calaun-learning/calaun

- Built and deployed a generative AI application pairing a symbolic math engine with a self-hosted Llama LLM for step-by-step solutions, serving 400+ weekly users with 90+ automated tests.

Large Language Model From Scratch github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Implemented a LLaMA-style transformer in PyTorch, including RMSNorm, SwiGLU, rotary embeddings, grouped-query attention, and a KV cache.

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes to show where LTV = ARPU / churn fails, estimating customer lifetime value from empirical survival curves.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, PyTorch), SQL, R, C#, Bash, PowerShell

Machine Learning & AI: Regression, Classification, Clustering, XGBoost, NLP, LLMs (Llama), Transformers, Uplift Modeling, Media Mix Modeling, Time Series Forecasting, Survival Analysis, Feature Engineering, Model Evaluation, Model Monitoring & Drift Detection

Statistics & Experimentation: Hypothesis Testing, A/B Testing, CUPED, Power Analysis, Causal Inference, Double ML, Causal Forests, Difference-in-Differences, CATE, Monte Carlo Simulation, Optimization

Data Engineering & Cloud: Spark, Databricks, Snowflake, AWS, Azure, SQL Server, PostgreSQL, Docker, Elasticsearch, ETL/ELT Pipelines, Data Quality Checks, Unit Testing, Git

Visualization & Communication: Power BI, Tableau, Excel, Matplotlib, Plotly, Dashboards, Stakeholder Presentations, Explaining Results to Non-Technical Audiences, Mentoring